

Supplementary Appendix

Low-Dimensional Generative Models of Human Interception Movements
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This appendix accompanies the eight-page scientific report. It supplies extended methods, complete comparisons and diagnostic results. Historical component outputs and matched-procedure refits are explicitly separated; neither is evidence of a validated cognitive strategy.

S1 Extended methods and reproducibility

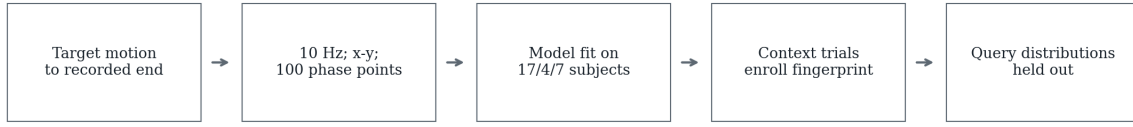
CVAE position input is 200 values; five condition values enter encoder and decoder. Physical timing is transformed as $\log(t+0.001)$, standardized with training statistics, and inverted as $\max(\exp(u)-0.001, 0)$. The loss sums 200 standardized coordinate errors and 20 times two timing errors, plus the KL penalty. Adam begins at 0.001; ReduceLROnPlateau halves learning rate after 15 validation plateaus. Early stopping uses patience 25 and at most 150 epochs; five of 96 neural histories hit the cap. Validation selection uses the configured variational validation objective, whereas reported reconstruction uses posterior means.

Spline fitting uses cubic B-splines with five interior knots and nine coefficients per coordinate (18 in total), followed by training-only PCA to the tested dimensions. Physical timings are excluded from spline-PCA inputs. Validation selects Ridge regularization for the timing head. The spline generator samples around context-mean coefficients with training-derived variation. Conditional AE removes stochastic latent sampling and KL; unconditional VAE suppresses condition inputs. Condition Ridge uses conditions alone and samples training residuals by task stratum.

A CVAE fingerprint is the context mean of posterior means. The shared latent covariance combines within-participant variation of training posterior means and average diagonal posterior uncertainty, plus a small diagonal regularizer. A shared latent covariance need not produce equal behavioural variances after nonlinear decoding. Mean-plus-SD fingerprints are separate higher-dimensional probe sensitivities.

The four outer folds test every person once (17/4/7 train/validation/test); neural seeds are 42, 43 and 44. Context/query splitting uses seed 2026 plus a deterministic subject offset, with approximately half the trials in each start-category/side stratum. Generated task draws use the target person's query condition metadata, not query shapes or timing labels. An enrolment score is classification among seven known context centroids after context-derived scaling; nominal chance is 1/7.

S1.1 Distance estimators and inference



Four outer participant folds; each participant tested once; neural seeds 42/43/44

Figure S1. Data, model and evaluation flow; the post-review controls extend the evaluation of frozen models.

Continuous multivariate features use training standard deviations, count scales are at least one count and constant scales are one. A training median-distance RBF bandwidth is fixed per fold using at most 512 examples. The unbiased squared-MMD U-statistic excludes within-sample diagonals and may be negative in finite samples. Energy is $2 \text{ mean}||X-Y|| - \text{mean}||X-X|| - \text{mean}||Y-Y||$ including diagonals and with no square root. Mean KS weights 11 feature statistics equally; it is not a combined hypothesis test. Heuristic speed peaks use 10% prominence and 50 ms separation; minimum-jerk counts use fitted components.

Main comparisons average seeds per participant, then apply two-sided Pratt Wilcoxon tests with 12-decimal difference rounding, SciPy 1.16.3 method='auto'. All 56 main significance decisions are unchanged by this numerical correction. BH families are separate: 56 main contrasts, eight timing sensitivities, 12 fingerprint controls and nine matched-component dimension comparisons. Other historical sensitivity families retain their recorded conventions. Cross-validation dependence limits inference. Core execution used Python 3.13, NumPy 2.3.4, pandas 2.3.3 and SciPy 1.16.3.

S2 Complete model comparison

Table S1. Complete dimension sweep; definitions and units follow main Table 2.

Model	n	MSE	Init. MAE (ms)	Move MAE (ms)	Mean KS	MMD ²	Enroll. (%)
Condition Ridge	-	1.234	98.5	72.7	0.277	0.126	-
Conditional AE	3	0.170	37.8	44.1	0.288	0.141	39.0
Conditional AE	8	0.026	39.9	42.7	0.259	0.111	46.5
CVAE	2	0.325	39.8	46.8	0.310	0.163	33.0
CVAE	3	0.177	37.8	43.4	0.278	0.131	40.9
CVAE	4	0.102	35.7	44.8	0.263	0.113	44.3
CVAE	8	0.032	34.8	44.2	0.219	0.067	50.7
Spline+PCA	2	0.197	56.6	50.9	0.355	0.214	29.6
Spline+PCA	3	0.097	56.2	51.2	0.325	0.181	36.4
Spline+PCA	4	0.076	56.2	51.2	0.302	0.167	38.4
Spline+PCA	8	0.023	55.6	50.2	0.255	0.115	48.6
Unconditional VAE	3	0.148	37.9	42.9	0.285	0.130	40.0
Unconditional VAE	8	0.033	36.0	42.2	0.215	0.064	50.8

Table S2. CVAE-spline contrasts within the 56-comparison BH family; timing values are ms. The complete 56-row CSV accompanies this appendix.

n	Endpoint	CVAE	Spline	CVAE lower /28	Adjusted p
3	Trajectory MSE	0.177	0.097	6	<0.0001
3	Movement MAE	43.446	51.160	24	<0.0001
3	Initiation MAE	37.832	56.155	27	<0.0001
3	Mean KS	0.278	0.325	26	<0.0001
3	MMD squared	0.131	0.181	24	<0.0001
8	Trajectory MSE	0.032	0.023	4	0.0008
8	Movement MAE	44.163	50.208	23	0.0014
8	Initiation MAE	34.837	55.584	27	<0.0001
8	Mean KS	0.219	0.255	23	0.0001
8	MMD squared	0.067	0.115	25	<0.0001

S3 Complete fingerprint controls

Table S3. Full same-decoder control family (12 tests). Wrong-person distances average the six other test-context donors before seed averaging.

n	Control	Endpoint	Control - own	Own lower /28	Adjusted p
3	population	Mean KS	0.033	22	0.0002
3	population	Energy	0.270	23	0.0002
3	population	MMD squared	0.044	23	0.0004
3	wrong	Mean KS	0.060	27	<0.0001
3	wrong	Energy	0.527	26	<0.0001
3	wrong	MMD squared	0.089	27	<0.0001
8	population	Mean KS	0.070	27	<0.0001
8	population	Energy	0.488	27	<0.0001
8	population	MMD squared	0.081	27	<0.0001
8	wrong	Mean KS	0.098	27	<0.0001
8	wrong	Energy	0.765	27	<0.0001
8	wrong	MMD squared	0.132	27	<0.0001

Every own-fingerprint result was checked against the existing saved distribution result before a control contrast was computed. Raw output retains each donor identity, context/query sample counts, per-feature distances and run identifiers. This control changes only the supplied latent centre; it does not retrain a model or use query trajectories to enrol a participant.

CVAE capacity and fingerprint enrollment

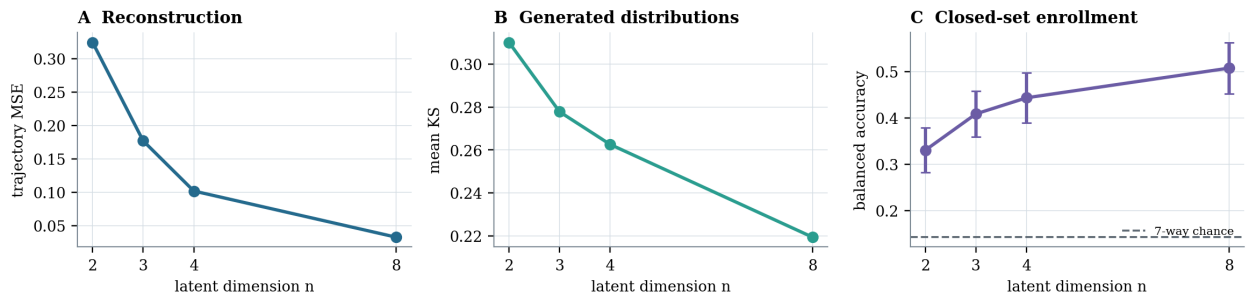


Figure S2. Capacity and enrolment across latent dimensions; error bars describe fold/seed spread, not independent-cohort confidence intervals.

S4 All behavioural prediction targets

Table S4. All 14 CVAE mean-fingerprint query-summary probes; each R-squared statistic is averaged over seeds. Mean-fold and pooled statistics are not interchangeable.

Query-summary target	n=3 pooled R^2	n=3 fold R^2	n=8 pooled R^2	n=8 fold R^2
Curvature index (mean)	-0.242	-5.660	-0.215	-5.672
Curvature index (sd)	-0.123	-1.204	-0.141	-1.499
Initiation time (mean)	0.723	0.661	0.707	0.655
Initiation time (sd)	-0.011	-0.188	-0.032	-0.803
Lateral deviation (mean)	0.060	-0.215	0.319	0.273
Lateral deviation (sd)	-0.426	-2.842	-0.214	-1.476
Movement time (mean)	0.026	-2.934	0.113	-2.337
Movement time (sd)	-0.109	-0.357	-0.105	-0.599
Speed-peak count (mean)	-0.305	-4.611	-0.267	-5.111
Speed-peak count (sd)	-0.354	-1.070	-0.257	-0.964
Path length (mean)	-0.388	-2.573	-0.130	-2.050
Path length (sd)	-0.099	-1.376	-0.114	-1.647
Peak speed (mean)	0.356	0.346	0.320	0.303
Peak speed (sd)	-0.103	-0.268	0.007	-0.135

Ridge regularization is chosen on validation participants, then the probe refits on the 21 development participants. The historical constant is based on the 17 training participants. The added constant instead uses the 21 development query-summary targets, matching access at probe refitting. Direct context copies each measured context mean or sample SD to predict the same quantity on query trials. It uses no query outcomes; taken jointly it retains 14 target-specific quantities rather than a shared n-dimensional code.

S4.1 Direct-context MAE on every target

Table S5. Mean absolute errors on all summary targets. Timing is ms; peak speed is tracker units/s; path length and lateral deviation use tracker units; curvature and counts are dimensionless. Descriptive comparison, without an additional hypothesis-test family.

Target	CVAE n=3	CVAE n=8	Direct context	21-person constant
Curvature index (mean)	0.016	0.016	0.008	0.016
Curvature index (sd)	0.054	0.055	0.039	0.054
Initiation time (mean) [ms]	14.778	14.711	9.275	30.381
Initiation time (sd) [ms]	13.717	14.449	10.928	14.028
Lateral deviation (mean)	0.188	0.159	0.038	0.214
Lateral deviation (sd)	0.112	0.100	0.067	0.104
Movement time (mean) [ms]	36.444	37.349	11.049	48.762
Movement time (sd) [ms]	30.951	32.617	27.879	32.441
Speed-peak count (mean)	0.068	0.066	0.022	0.066
Speed-peak count (sd)	0.139	0.134	0.079	0.134
Path length (mean)	0.386	0.330	0.162	0.369
Path length (sd)	0.677	0.689	0.438	0.673
Peak speed (mean)	7.744	8.763	1.734	11.876
Peak speed (sd)	2.627	2.441	1.707	2.646

The direct-context baseline has lower MAE than the CVAE probe on all 14 targets at each dimension. This does not establish equivalence of information budgets or superiority for full trajectory generation. All model-family probe results and mean-plus-SD variants remain available in the machine-readable tables.

S4.2 Timing calibration details

The common timing MLP has two 64-unit hidden layers, participant weighting, training-only input/log-target scaling, validation early stopping (patience 40) and at most 400 epochs. There are 48 such fits. The multiplicative calibration is positive and selected using validation MAE for each timing endpoint; it does not alter the main trajectory generator. Main Table 4 includes all eight paired comparisons.

S5 Component refitting and sensitivity

Table S6. Matched-procedure distances and empirical sampling reference; repeated here beside diagnostics.

Component discrepancy	n=3	n=8	Sampling reference
Fit error KS	0.339	0.308	0.266
First duration KS	0.455	0.409	0.267
First displacement amplitude KS	0.377	0.351	0.267
Secondary amplitude fraction KS	0.287	0.286	0.197
Overlap KS	0.267	0.268	0.195
Count total variation	0.214	0.232	0.146
Count JSD	0.076	0.087	0.047

Table S7. Optimizer diagnostics. A completed procedure can include nonconverged candidate orders; convergence is not a guarantee of a global optimum.

Input	Fits	Selected converged	All candidates converged	Legacy count agreement
generated	1680	1680	1440	99.7%
recorded	2376	2376	2111	76.1%

The component basis is $(30u \text{ squared} - 60u \text{ cubed} + 30u \text{ to the fourth})/\text{duration}$, multiplied by x/y displacement and zero outside the component interval. The objective includes x/y residuals and the difference between norms of summed velocities. It pads the observation with zeros beyond the recorded interval. Bounds allow durations from 0.100 to 1.000 s; the 0.050 s setting is an indexed absolute onset lower-bound step, not a pairwise onset gap. The BIC calculation includes correlated speed and zero padding, so it is a model-order heuristic here.

Recorded inputs in this rerun are the canonical filtered 100-point go-window targets; generated inputs are decoder outputs on the same grid. Both use the same timing-based crop, low-pass filter and two-restart/400-evaluation fitting code. Physical movement timing is measured for recordings and predicted for generated samples. The pre-acquisition/target filtering history is not erased. This control therefore evaluates the represented trajectories with a common downstream procedure; it does not validate the inferred components as distinct biological commands.

S5.1 Dimension comparisons and optimizer sensitivity

Table S8. Nine paired matched-component dimension contrasts, with one BH family.

Endpoint	n=3	n=8	n=8 lower /28	Adjusted p
KS fit error	0.339	0.308	16	0.2626
KS first duration s	0.455	0.409	18	0.2626
KS first amplitude	0.377	0.351	17	0.2626
KS secondary amplitude fraction	0.287	0.286	14	0.8109
KS mean overlap pct	0.267	0.268	10	0.6448
Count TV	0.214	0.232	10	0.3478
Count JSD	0.076	0.087	11	0.3478
BIC count TV	0.179	0.158	12	0.6840
BIC count JSD	0.059	0.043	17	0.3478

Table S9. First two query trials and first two generated samples per person, n=3/8 seed42; four restarts and 700 evaluations compared with the matched base.

Input	Subset n	Count agreement	Base selected converged	High-budget converged
generated	112	99.1%	112	112
recorded	56	100.0%	56	56

Historical count TV was 0.353/0.355 at n=3/8. Those recorded fits used two restarts and 400 evaluations on already-filtered 240-Hz movement slices; generated fits used one restart and 300 evaluations. On 56 historical recorded trials, four-restart/700-evaluation counts agreed in 100%, while 167 ms duration/onset-bound and 5 Hz changes agreed in 55.4% and 51.8%. These historical sensitivities do not certify the new procedure; they show why fitting assumptions must remain visible.

The matched sampling reference resamples each observed query component distribution 500 times: an independent query-sized and ten-sample draw, with three seed-like replicates averaged before equal-person aggregation. It conditions on the fitted empirical distribution and does not include fitting or training uncertainty. No reference value is a hard threshold for a satisfactory model.

S6 Conditioning and latent associations

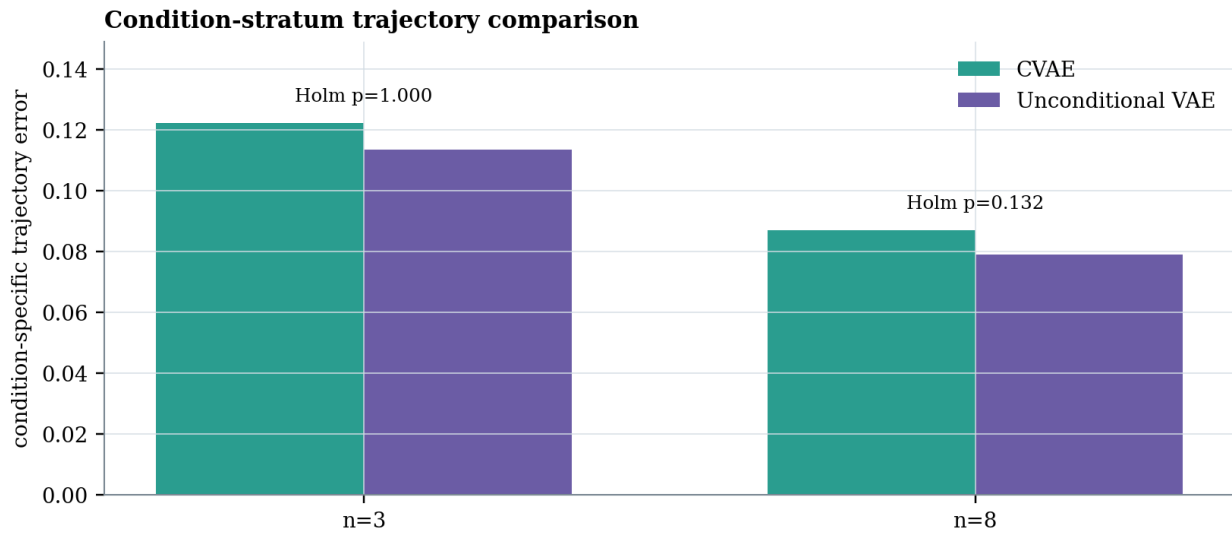


Figure S3. Condition-stratum trajectory diagnostic. Holm-adjusted $p=1.000$ ($n=3$) and 0.132 ($n=8$); no feature-level contrast among 44 survives BH correction.

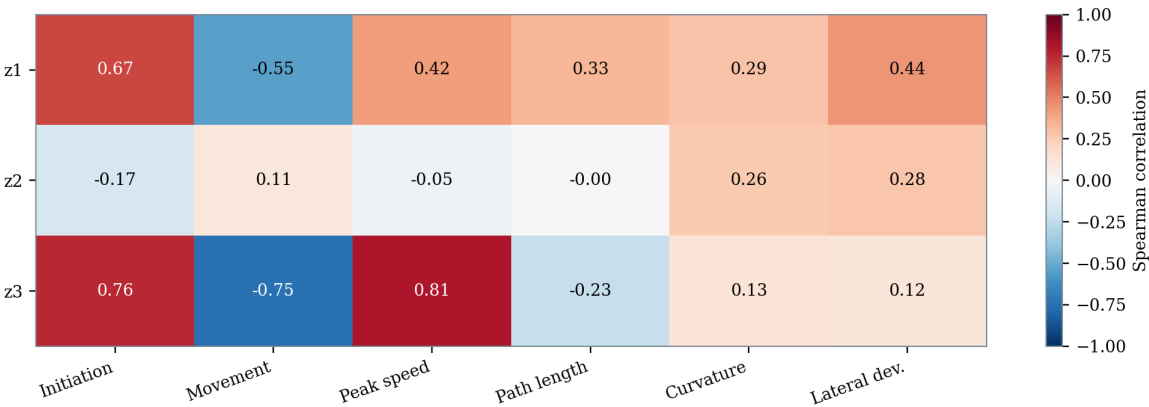


Figure S4. Within-decoder Spearman correlations for 500 training-centred latent draws at fixed $sp=2$, $side=1$ and speed 0.635 screen widths/s; CVAE $n=3$, fold0, seed42. Correlated draws and coordinate non-identifiability limit interpretation.

K-means on the available normalized trajectories is an in-sample description (ARI 0.052; permutation $p=0.00498$ with 200 permutations). It is not an input to the CVAE, a held-out identity classifier or evidence that individual latent coordinates have a cognitive meaning. Query enrolment, personal generation and condition manipulation are separate evaluations.

S7 Representative trajectories and event audit

Held-out reconstruction and context generation (fold 0, seed 42)

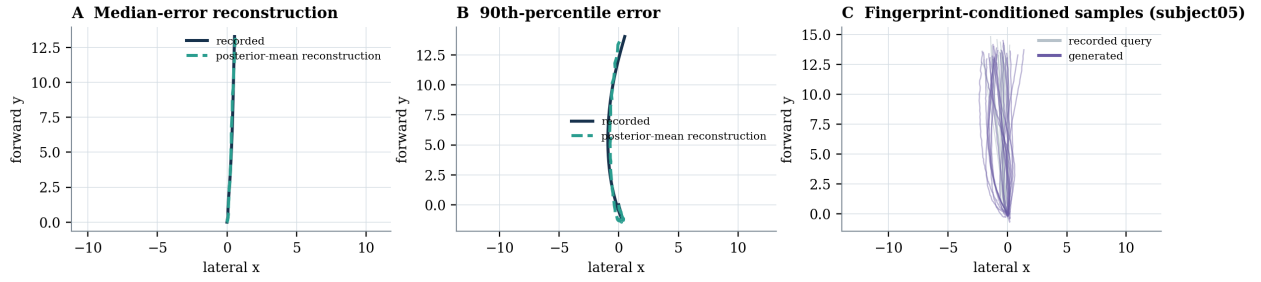


Figure S5. Held-out posterior-mean reconstructions and context generation, CVAE $n=8/\text{fold}0/\text{seed}42$. Typical trial subject13_li_2_1_2_7; high-error trial subject19_li_2_1_1_8. Selection uses reconstruction-error quantiles, not a claim that these examples summarize all participants.

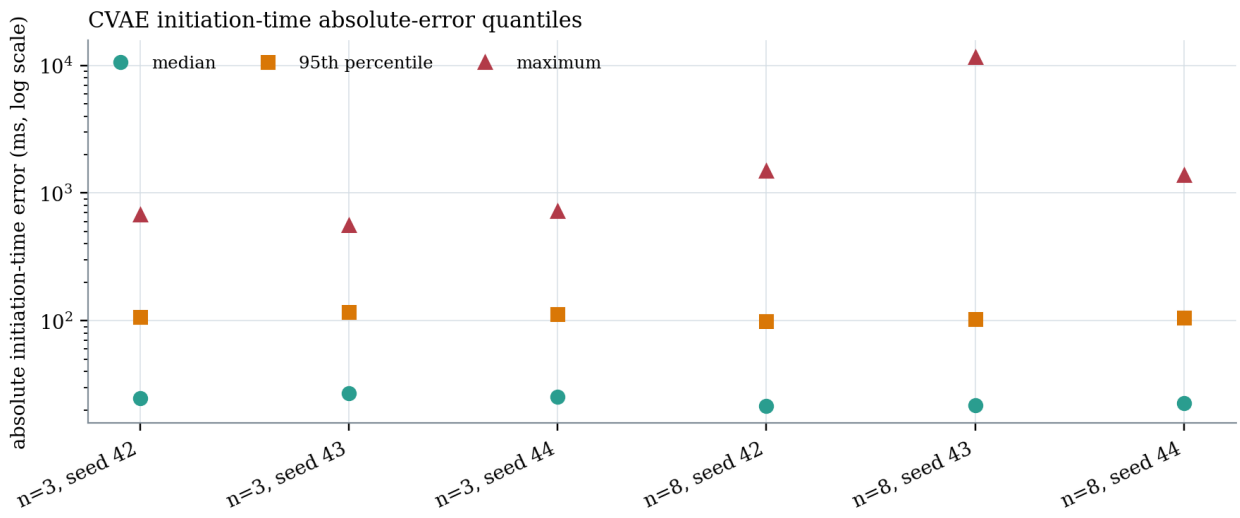


Figure S6. Original initiation-time absolute-error quantiles and maxima. Small typical errors coexist with extreme failures; all remain in the reported benchmark.

S7.1 Early-movement examples

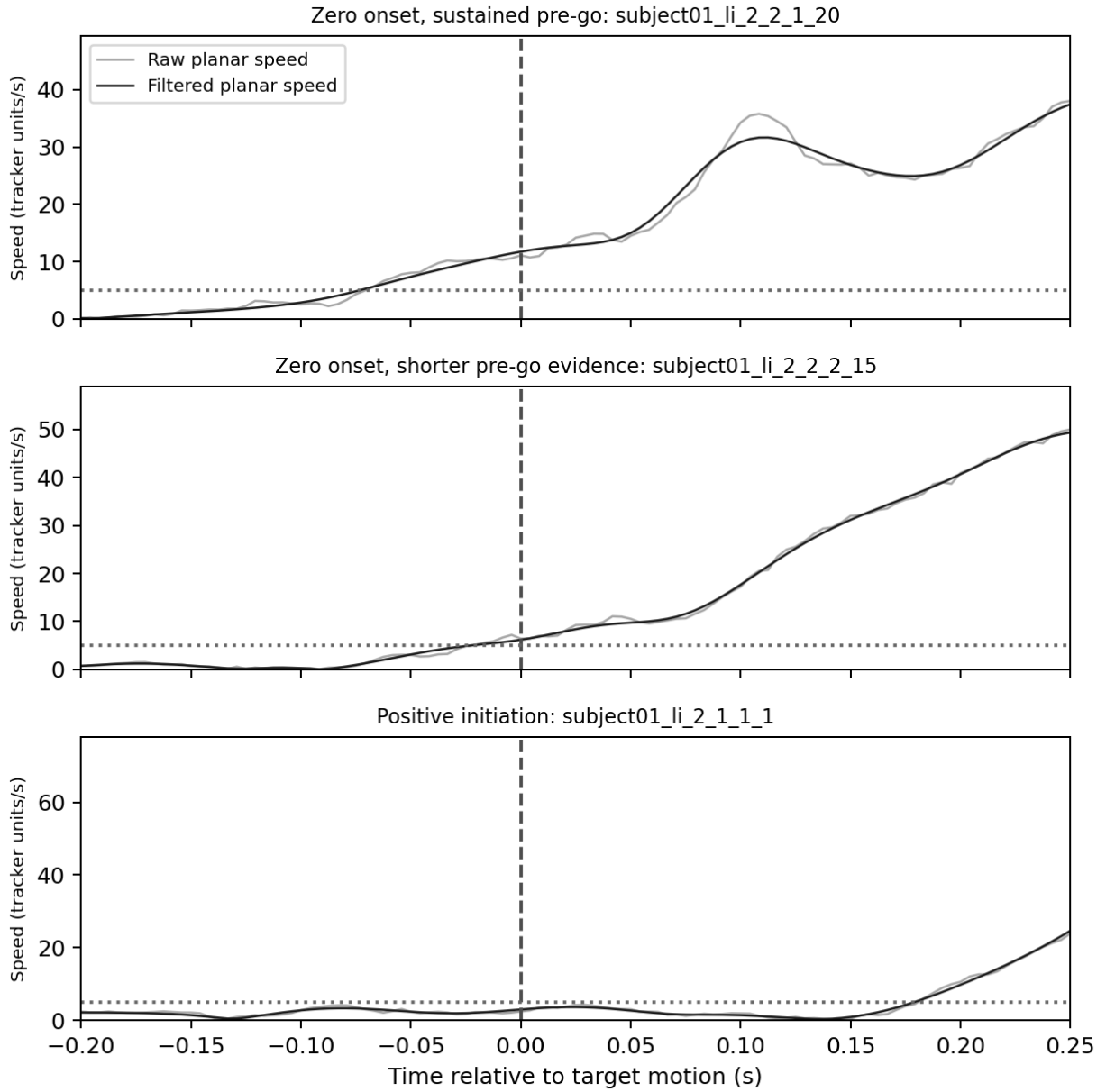


Figure S7. Deterministically selected examples from the event-audit groups. The full table contains every retained trial; threshold exceedance is not a diagnosis of premature movement or sensor noise.

The 357 zero-initiation cases all satisfy the immediate sustained threshold; none uses the missing-onset fallback. The 109 filtered and 100 raw pre-target-motion exceedances use a 50 ms planar-speed criterion. They are retained; the current encoder window begins at target motion. The 26.2 ms mean CSV/MAT endpoint offset is a measured proxy difference, not a demonstrated synchronization correction.

S8 Artifacts and execution

The advisor package contains the eight-page report, this appendix, the student guide, four illustrative CVAE checkpoints, a Streamlit dashboard and machine-readable summaries. Raw Dropbox recordings are not bundled. The dashboard's live generation uses fold0/seed42 and dimensions 2,3,4,8; benchmark tables use the declared full matrix. Condition controls are exploratory.

To launch the bundled dashboard, install requirements_dashboard.txt and run `python -m streamlit run src/confirmatory_dashboard.py` from the extracted folder. The full research repository is needed to rerun evaluation scripts against the canonical cache and all saved checkpoints.

Table S10. Directories under studies/review_corrected_evaluation/results. Historical checkpoints and component fits remain under studies/final_strategy_evaluation.

Result directory	Contents
analysis	Main 56 paired contrasts, full model sweep, per-feature fidelity and timing tails
behavioral_probes	All targets, model families, mean/mean-plus-SD probes and out-of-fold predictions
timing_fairness	Common-head and calibrated timing predictions, factors and comparisons
event_audit	Trial-level event measurements and raw/filtered examples
review_controls	Fingerprint donor controls, direct-context baselines, matched components, convergence and sensitivities
sampling_reference	Historical empirical component-sampling reference

The post-review runners are scripts/run_review_controls.py (fingerprint or components) and scripts/analyze_review_controls.py (context or components). The component runner records resumable job IDs. Frozen settings and checkpoint hashes live beside the new results. Tests verify identical fitted results for identical recorded/generated inputs and check physical-time restoration. Own-fingerprint generation is checked against every original saved n=3/8 fold/seed benchmark.

Remaining domain questions

Confirm permitted movement timing and tracker/MAT alignment, the endpoint proxy, fixation-warning inclusion, late-arrival rule, tracker units and onset threshold. Confirm whether an earlier appearance-to-end model window is scientifically required before changing the training representation. The current protocol preserves all early-motion recordings and reports its window explicitly.